Blacksburg, VA
Pratheek Prakash Shetty
+1 (540)-934-8350
pratheekps@vt.edu
LinkedIn: pratheekps01

"Fullstack engineer with experience in backend and ML, specializing in performance, parallel processing, and scaling."

SKILLS

- Proficient in: Python, C++, C, Java, JavaScript, Rust, Go, TypeScript, SQL, RabbitMq, gRPC, kafka
- Parallel Computing: CUDA, OpenMP, Multithreading, High-Performance Computing (HPC)
- Infrastructure: Docker, Kubernetes, microservices

EDUCATION

Master of Science, Computer Engineering Aug 2024 - May 2026 (Expected)
Virginia Polytechnic Institute and State University

Bachelor of Engineering, Computer Science and Engineering Aug 2019 - Aug 2023
Acharya Institute of Technology, Bengaluru

PROFESSIONAL EXPERIENCE

Software Engineer, Virginia Tech Sep 2024 - Present
- Building a high-speed data acquisition GUI using Python backend and JavaScript with Electron and React.
- Developed a software serial interface for real-time hardware communication.
- Processed and visualized large datasets using Python scientific libraries.

SDE Intern, Instahyre Mar 2024 - Jun 2024
- Contributed to Full Stack engineering with CI/CD and microservices architecture with Django backend.
- Worked with and developed Vuejs frontend involving the application hiring process.
- Developed customizable job email templates using Django, Angular, and RabbitMQ.

SDE Intern, Tikaj Private Ltd Mar 2023 - Jul 2023
- Maintained and enhanced Node.js backend for Phishgrid and Blinkstore, integrating key features.
- Integrated an LLM into Blinkstore, building scripts to invoke OpenAI models for automated task execution.
- Worked on CI/CD pipelines and microservices architecture to improve backend scalability.

Backend Intern, CryptoNaukri Apr 2022 - Jun 2022
- Designed and implemented secure API routes for backend using Express.js and express-validator.
- Improvednode.js backend performance by 10% through secure and validated route optimization.
- Contributed to the development of a community platform for blockchain developers.

PROJECTS

GigaDAQ: Real-Time Data Acquisition System Sep 2024 - Present
- Developing a Python server for real-time data acquisition from custom lab hardware.
- Building an Electron frontend for dynamic visualization with waveforms and plots.
- Optimizing server performance with low-level data handling and network I/O improvements.

AIT, Iridescence.ai: FREENeRF Enhanced Text-to-3D Mesh Models Sep 2022 - Jun 2023
- Led a team to optimize 3D mesh model generation from text prompts using a modified DreamFusion pipeline.
- Improved NeRF efficiency with FREENeRF for few-shot learning, enhancing model accuracy.

Racing F1 Manager *Jan 2022 - Feb 2022*
- Developed a sports management tool for F1 teams using TypeScript, GraphQL, MySQL, and React.
- Unified team management into a single portal with positive user feedback.